

- b) Write an assembly language program in 8086 to count the number of O's in the word "Information Technology". 8

7. Write short notes on: (**Any two**)

2×5

- a) Interrupt controller
- b) Unique address decoding
- c) DAA Instruction with Example

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2021

Programme: BE

Full Marks: 100

Course: Microprocessor and Assembly Language
Programming

Pass Marks: 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define microprocessor with examples. Differentiate between Von Newmann and Harvard architecture along with its block diagram. 7
b) What is flag? Explain the main components of 8085 microprocessor with neat and clean 8085 architecture. 8
2. a) What is RTL? Draw the timing diagram of ADI 23 H and calculate the total execution time if $f = 3\text{Mhz}$. 7
b) What are the signals of timing & control block of 8085 microprocessor? Explain with its functionality. 8
3. a) Write an assembly language program for 8086 to find the factorial of the number entered by user in Dos screen. 7
b) Describe stack and subroutine with examples. Write an assembly language program to count the prime numbers from the 10 data's stored in memory started from C060H . Also stored the count in C090H. 8
4. a) Explain the READ and WRITE operation of Dynamic RAM cell along with its diagram. 7
b) Design an address Decoding circuit to interface two RAM blocks and a ROM block each of 4KB starting at address 4000H. 8
5. a) Explain the flow of interrupt processing in detail. What is the difference between polled and Daisy chained multiple interrupt handling technique? 7
b) Describe the different modes of 8255 A PPI with block diagram. Write the control word to set PC3. 8
6. a) What is DMA? Describe the purpose of DMA and explain the data transfer from input-output through DMA to memory with neat and clean diagram. 7